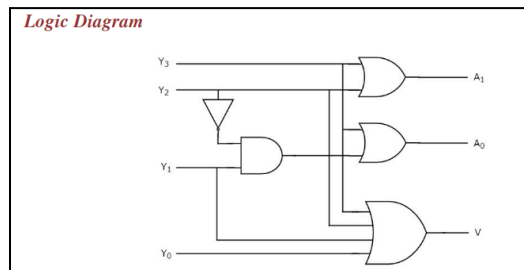


# Circuit Simulation Lab: 5

The purpose of this experiment is to introduce the design of simple combinational circuits, in this case Priority Encoder. A 4 to 2 priority encoder has four inputs  $Y_3, Y_2, Y_1$  &  $Y_0$  and two outputs  $A_1$  &  $A_0$ . Here, the input,  $Y_3$  has the highest priority, whereas the input,  $Y_0$  has the lowest priority. In this case, even if more than one input is '1' at the same time, the output will be the binary code corresponding to the input, which is having higher priority. We considered one more output,  $V$  in order to know, whether the code available at outputs is valid or not. If at least one input of the encoder is '1', then the code available at outputs is a valid one. In this case, the output,  $V$  will be equal to 1. If all the inputs of encoder are '0', then the code available at outputs is not a valid one. In this case, the output,  $V$  will be equal to 0

Software tools Requirement

ICARUS & GTKWAVE



| 4 to 2 Priority Encoder |       |       |       |         |       |     |
|-------------------------|-------|-------|-------|---------|-------|-----|
| Inputs                  |       |       |       | Outputs |       |     |
| $Y_3$                   | $Y_2$ | $Y_1$ | $Y_0$ | $A_1$   | $A_0$ | $V$ |
| 0                       | 0     | 0     | 0     | 0       | 0     | 0   |
| 0                       | 0     | 0     | 1     | 0       | 0     | 1   |
| 0                       | 0     | 1     | x     | 0       | 1     | 1   |
| 0                       | 1     | x     | x     | 1       | 0     | 1   |
| 1                       | x     | x     | x     | 1       | 1     | 1   |

DATAFLOW

```
module df_prio (
input [3:0] y ,
output [1:0] a ,
output v );

assign a[1] = y[3] | (~y[2] & y[1] );
assign a[0] = y[3] | y[2] ;
assign v = y[3] | y[2] | y[1] | y[0] ;

endmodule
```

TESTBENCH

```

module df_tb_prio;
reg [3:0] y ;
wire [1:0] a ;
wire v ;

df_prio p1 (
.y(y),.a(a),.v(v));

initial begin

$dumpfile ("df_prio.vcd");
$dumpvars (0,df_tb_prio);

repeat (20) begin
y = $random %16 ;
#20 ;
$display ("y = %b , a = %b , v = %b ", y , a , v );
end

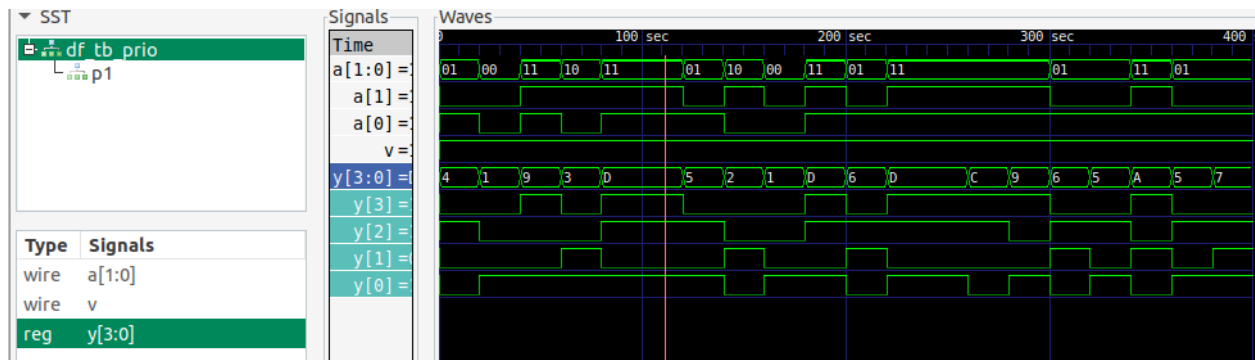
end
endmodule

```

```

vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_5$ vvp out
VCD info: dumpfile df_prio.vcd opened for output.
y = 0100 , a = 01 , v = 1
y = 0001 , a = 00 , v = 1
y = 1001 , a = 11 , v = 1
y = 0011 , a = 10 , v = 1
y = 1101 , a = 11 , v = 1
y = 1101 , a = 11 , v = 1
y = 0101 , a = 01 , v = 1
y = 0010 , a = 10 , v = 1
y = 0001 , a = 00 , v = 1
y = 1101 , a = 11 , v = 1
y = 0110 , a = 01 , v = 1
y = 1101 , a = 11 , v = 1
y = 1101 , a = 11 , v = 1
y = 1101 , a = 11 , v = 1
y = 1100 , a = 11 , v = 1
y = 1001 , a = 11 , v = 1
y = 0110 , a = 01 , v = 1
y = 0101 , a = 01 , v = 1
y = 1010 , a = 11 , v = 1
y = 0101 , a = 01 , v = 1
y = 0111 , a = 01 , v = 1

```



## BEHAVIOURAL

```
module bm_prio (
input [3:0] y ,
output reg [1:0] a ,
output reg v );
```

```
always @(*) begin
```

```
if (y[3] == 1 ) begin
a = 2'b11;
v = 1 ;
end
```

```
else if (y[2] == 1 ) begin
a = 2'b10;
v = 1 ;
end
```

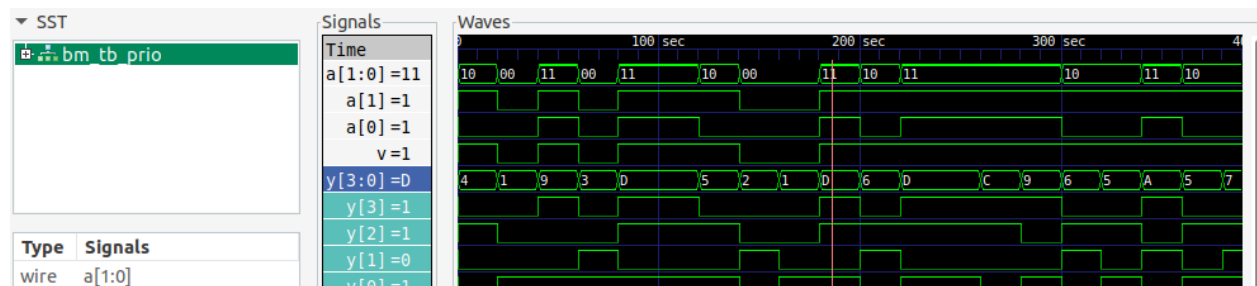
```
else if (y[3] == 1 ) begin
a = 2'b01;
v = 1 ;
end
```

```
else if (y[3] == 1 ) begin
a = 2'b00;
v = 1 ;
end
```

```
else begin
v = 0 ;
a = 2'b00;
```

```
end
end
endmodule
```

```
VCD info: dumpfile bm_prio.vcd opened for output.
y = 0100 , a = 10 , v = 1
y = 0001 , a = 00 , v = 0
y = 1001 , a = 11 , v = 1
y = 0011 , a = 00 , v = 0
y = 1101 , a = 11 , v = 1
y = 1101 , a = 11 , v = 1
y = 0101 , a = 10 , v = 1
y = 0010 , a = 00 , v = 0
y = 0001 , a = 00 , v = 0
y = 1101 , a = 11 , v = 1
y = 0110 , a = 10 , v = 1
y = 1101 , a = 11 , v = 1
y = 1101 , a = 11 , v = 1
y = 1100 , a = 11 , v = 1
y = 1001 , a = 11 , v = 1
y = 0110 , a = 10 , v = 1
y = 0101 , a = 10 , v = 1
y = 1010 , a = 11 , v = 1
y = 0101 , a = 10 , v = 1
y = 0111 , a = 10 , v = 1
```



## DATAFLOW

```
module sm_prio(
input [3:0] y ,
output [1:0] a ,
output v );
```

```
wire not_y2;  
    wire and_out;
```

```
not gate_not (not_y2, y[2]);
```

```
and gate_and (and_out, not_y2, y[1]);
```

```
or gate_a1 (a[1], y[3], and_out);
```

```
or gate_a0 (a[0], y[3], y[2]);
```

```
or gate_v (v, y[3], y[2], y[1], y[0]);
```

```
endmodule
```

```
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_5$ vvp out
VCD info: dumpfile sm_prio.vcd opened for output.
y = 0100 , a = 01 , v = 1
y = 0001 , a = 00 , v = 1
y = 1001 , a = 11 , v = 1
y = 0011 , a = 10 , v = 1
y = 1101 , a = 11 , v = 1
y = 1101 , a = 11 , v = 1
y = 0101 , a = 01 , v = 1
y = 0010 , a = 10 , v = 1
y = 0001 , a = 00 , v = 1
y = 1101 , a = 11 , v = 1
y = 0110 , a = 01 , v = 1
```

